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$$\begin{aligned} & \int \tan^{-3} x dx \\ &= \int \cot^3 x dx \\ &= \int \cot x \cdot \cot^2 x dx \\ &= \int \cot x \left(\frac{1}{\sin^2 x} - 1 \right) dx \\ &= \int \cot x \frac{1}{\sin^2 x} dx - \int \cot x dx \\ &u = \cot x \Rightarrow du = -\frac{1}{\sin^2 x} dx \\ &= - \int u du - \ln |\sin x| \\ &= -\frac{u^2}{2} - \ln |\sin x| + C \\ &= -\frac{\cot^2 x}{2} - \ln |\sin x| + C \end{aligned}$$

References